

V8 Engine Assembly

Cylinder Block Assembly Guide

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CONFIDENTIAL

Torque Specifications

Component	Torque (Nm)	Grade
Main Bearing Cap Bolt	65 +/- 3	A
Connecting Rod Bolt	45 +/- 2	A
Flywheel Bolt	90 +/- 5	B
Oil Pan Bolt	25 +/- 2	C
Timing Cover Bolt	15 +/- 1	C
Cylinder Head Bolt (Stage 1)	40	A
Cylinder Head Bolt (Stage 2)	80	A
Cylinder Head Bolt (Final)	120	A

Note: All torque values must be verified with calibrated torque wrench.

Assembly Procedure

- Step 1: Clean all mating surfaces thoroughly
- Step 2: Apply assembly lubricant to bearing surfaces
- Step 3: Install main bearing shells (check oil holes alignment)
- Step 4: Position crankshaft carefully into bearing saddles
- Step 5: Install main bearing caps (numbered sequence)
- Step 6: Torque main bearing cap bolts to specification
- Step 7: Check crankshaft end play (0.05-0.25mm)
- Step 8: Install piston/connecting rod assemblies
- Step 9: Torque connecting rod bolts to specification
- Step 10: Verify crankshaft rotates freely by hand

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